

Detection of diabetic retinopathy using a fusion of textural and ridgelet features of retinal images and sequential minimal optimization classifier

[Ramasamy LK et. al, PeerJ Comput Sci]



Abstract

Diabetes is one of the most prevalent diseases in the world, which is a metabolic disorder characterized by high blood sugar. Diabetes complications are leading to Diabetic Retinopathy (DR). The early stages of DR may have either no sign or cause minor vision problems, but later stages of the disease can lead to blindness. DR diagnosis is an exceedingly difficult task because of changes in the retina during the disease stages.

An automatic DR early detection method can save a patient's vision and can also support the ophthalmologists in DR screening. This paper develops a model for the diagnostics of DR using a fusion of textural and ridgelet features of retinal images and sequential minimal optimization classifier

Objective

- Extract and fuse the ophthalmoscopic features from the retina images based on textural gray-level features.
- Apply texture analysis method like GLCM, GLRLM and Ridgelet Transform for feature extraction.
- Implement Sequential Minimal Optimization (SMO) for diabetic retinopathy recognition.
- Evaluate the performance of these methodology on two different DR dataset.
- Compare the performance evaluation result with previous related works.

Dataset

DIARETDB1 (DIARETDB1,2007) and KAGGLE(Kaggle dataset,2019) dataset used in this research.

Dataset	Dataset Description
DIARETDB1	• contains 89 fundus color images (size of 1500 x 1152). • 84 images are assigned as DR, and the remaining 5 images are normal.
Kaggle	• Contains 3662 retina scan images having a size of 224 x 224 pixels. • This dataset contains 5 categories of colored fundus images: No DR, Mild, Moderate, Severe, and PDR.

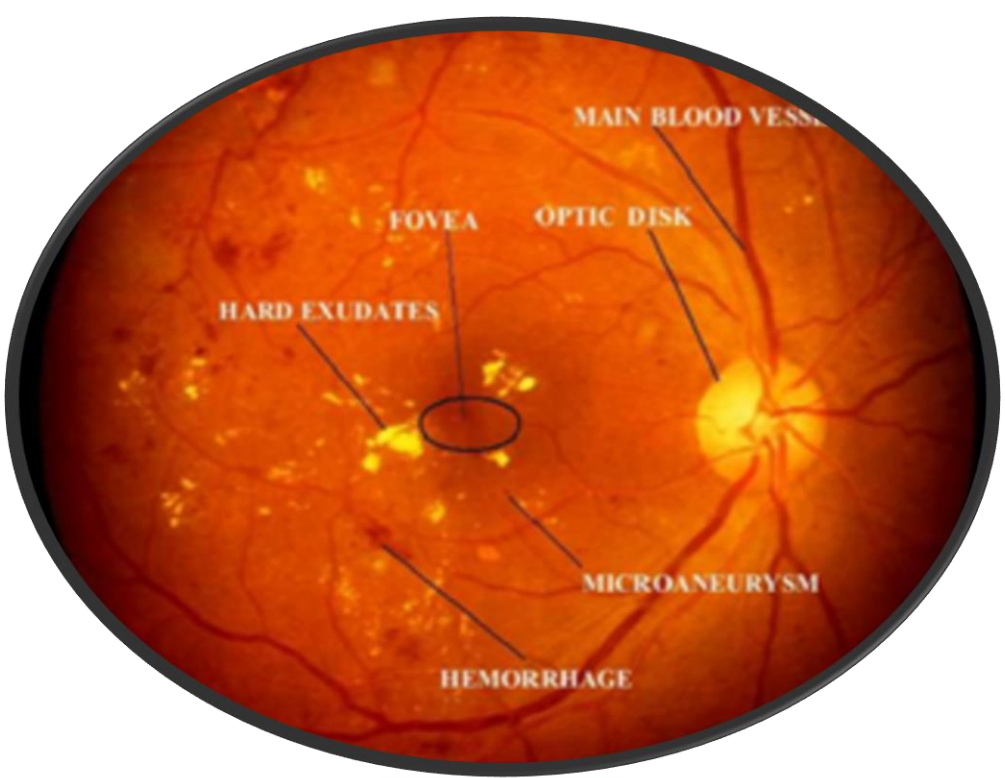
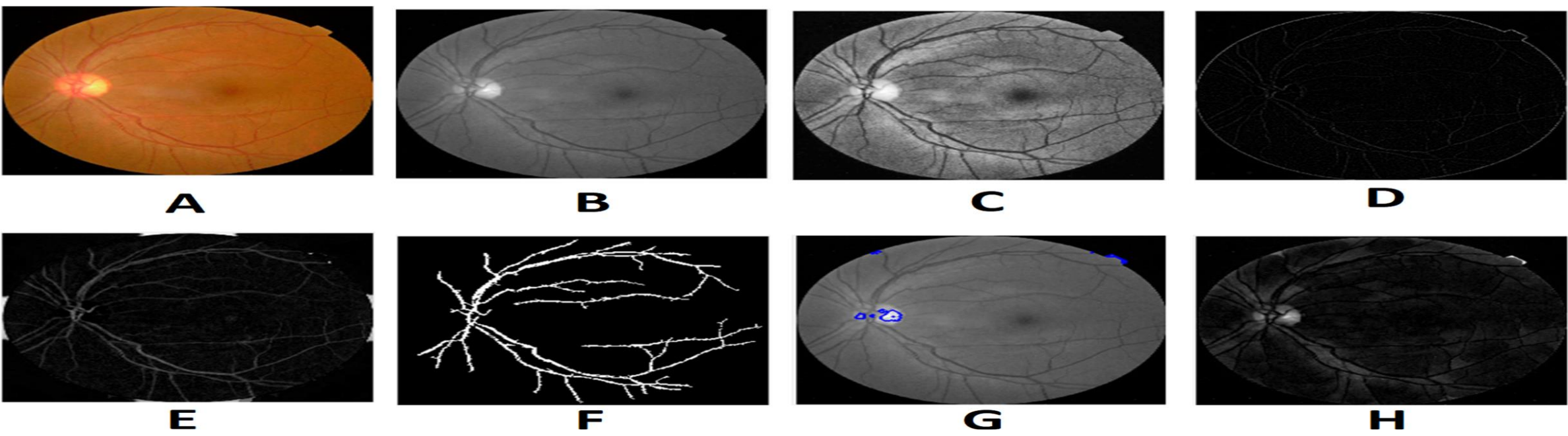


Fig: Illustration of retinal image features. Image credit: DIARETDB1

Methodology

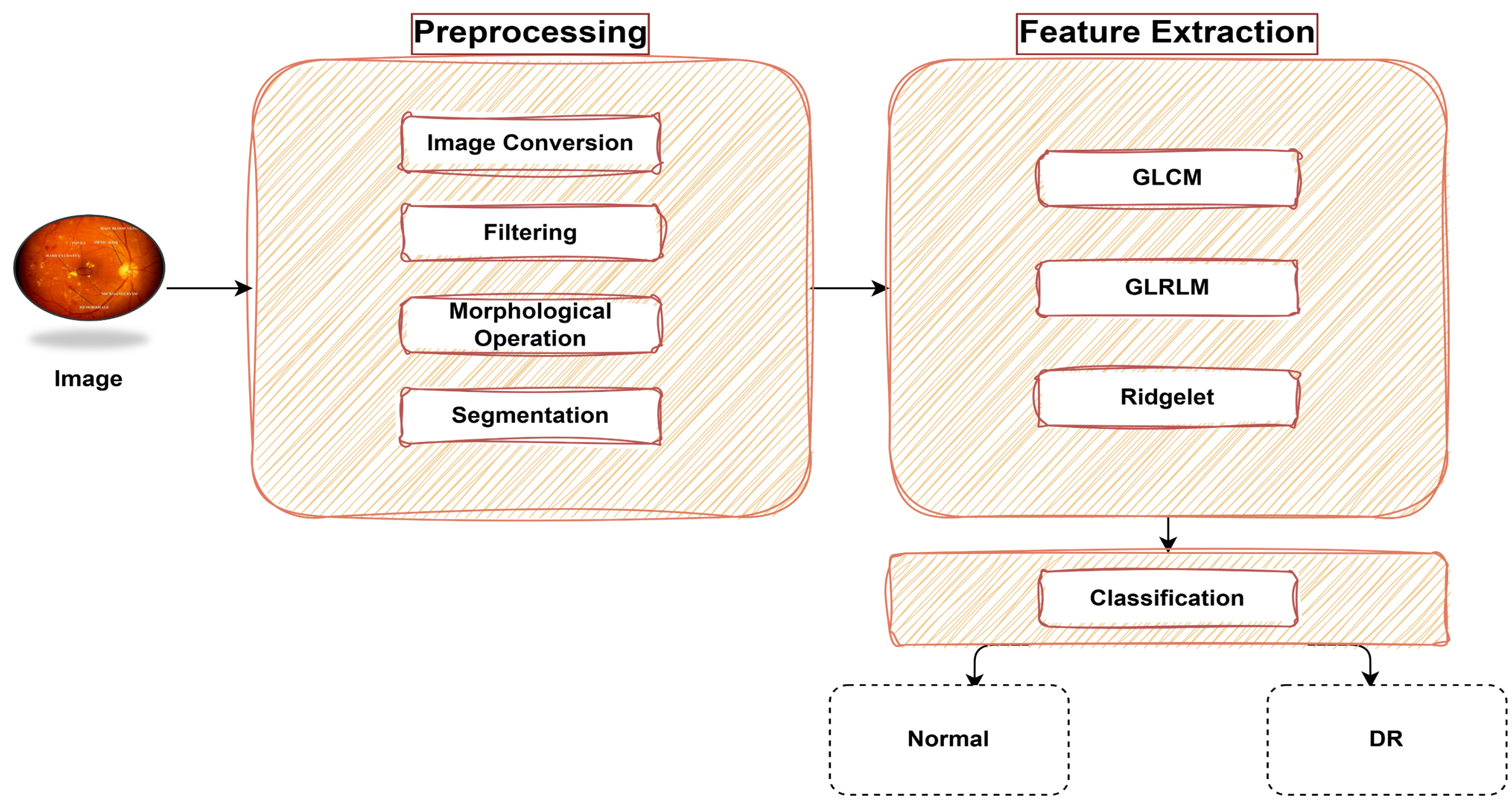
Preprocessing Stage

- **Image Conversion:** Retina image divided to RGB channel.
- **Filtering:** Laplacian filter used for denoising the retina image.
- **Morphological Operation:** Bottom hat transform, and Top hat transform applied to improve contrast and locate size pits in the grayscale image.
- **Segmentation:** Binarization is used to identify blood vessels and regions of the candidate (1), and other background regions (0).



Preprocessed result of normal retina image:

- (A) input image
- (B) green channel
- (C) histogram enhanced
- (D) filtered image
- (E) after bottom hat transform
- (F) after top hat Transform
- (G) blood vessels segmented
- (H) contours enhanced.



Workflow Diagram

Feature Extraction

Gray Level Co-occurrence Matrix (GLCM): GLCM focuses on the spatial relationships between pixel intensities in an image. It measures how often pairs of specific gray levels occur together at certain distances and angles within an image.

Gray Level Run Length Matrix (GLRLM): GLRLM characterizes the lengths of consecutive runs of pixels with the same gray level. It is useful for capturing information about the size and continuity of specific features within an image, such as lines or textures.

Continuous ridgelet transform (CRT): CRT decomposes an image or signal into a set of ridge-like features, which are elongated and oriented in specific directions. These ridgelets are particularly effective in capturing information along these preferred directions. The idea of the latter is to display linear features image to point using Radon transform and the subsequent use of wavelet transformations.

Classification

Sequential Minimal Optimization (SMO)

- SMO is a straightforward algorithm that uses only two Lagrange multipliers at each iteration to move the chunking process to the nearest possible expression.
- It determines the optimal value for these multipliers and updates the Support Vector Machin (SVM) until it fixes the whole QP problem.
- The benefit of SMO is that the optimization subproblem can be solved analytically with two Lagrange multipliers.

Results

Performance Metrics

$$Sensitivity (SE) = \frac{TP}{TP + FN} * 100\%$$

$$Specificity (SP) = \frac{TN}{TN + FP} * 100\%$$

$$Accuracy (ACC) = \frac{TP + TN}{N} * 100\%$$

$$F - score = \frac{2TP}{2TP + FP + FN}$$

TP	Count of true positive class (normal retina)
TN	Count of true negative class (DR)
FP	Count of false-positive (normal retina predicted as DR)
FN	Count of false-negative (DR predicted as Normal Retina)

Dataset	Feature Extraction Time (s)
DIARETDB1	131.56
KAGGLE	159.19

Dataset	SE (%)	SP (%)	ACC (%)	F-score
DIARETDB1	98.87	95.24	97.05	0.969
KAGGLE	90.9	91.0	91.0	0.909

Table: Comparison of performance evaluation results

Ref	Method	SE	SP	ACC	Dataset
[2] (Long et al., 2019)	Fuzzy C-means clustering (FCM) & support vector machine (SVM)	97.5	97.8	97.7	DIARETDB1
[3] (Chetoui & Akhloufi, 2020)	Extended Inception-Resnet-v2 network fine-tuned by cosine annealing strategy	98.8	90.1	97.1	DIARETDB1
[4] (Alaguselvi & Murugan, 2020)	Morphological operation, matched filter, principal component analysis (PCA), edge detection by ISODATA, and convex hull transform	99.03	98.37	98.68	DIARETDB1
Proposed	A fusion of texture and ridgelet features & SMO	98.7	95.24	97.05	DIARETDB1
[5] (Sudha & Ganeshbabu, 2021)	VGG-19 model, structure tensor and active contour approximation	98.83	96.76	98.28	Kaggle
[6] (Vaishnavi, Ravi & Anbarasi, 2020)	Contrast-limited adaptive histogram equalization (CLAHE) model and AlexNet architecture with SoftMax layer	92.00	97.86	95.86	Kaggle
Proposed	A fusion of texture and ridgelet features & SMO	98.7	95.24	97.05	Kaggle

Conclusion

The integration of the extracted features using texture analysis methods (GLCM and GLRLM) and Ridgelet Transform features suggests an automated approach for classifying Diabetic Retinopathy (DR). The extracted features using the suggested approach are used for the process of classification using the SMO classifier to identify DR. The results show that the proposed method is competitive with other state-of-the-art methods on the DIARETDB1 and KAGGLE datasets

References

- [1] Ramasamy LK, Padinjappurathu SG, Kadry S, Damaševičius R. 2021. Detection of diabetic retinopathy using a fusion of textural and ridgelet features of retinal images and sequential minimal optimization classifier. PeerJ Computer Science 7:e456
- [2] Shengchun Long, Xiaoxiao Huang, Zhiqing Chen, Shahina Pardhan, Dingchang Zheng, "Automatic Detection of Hard Exudates in Color Retinal Images Using Dynamic Threshold and SVM Classification: Algorithm Development and Evaluation", BioMed Research International
- [3] Mohamed Chetoui and Moulay A. Akhloufi "Explainable end-to-end deep learning for diabetic retinopathy detection across multiple datasets," Journal of Medical Imaging 7(4), 044503 (28 August 2020).
- [4] Alaguselvi R, Murugan K. 2020. Performance analysis of automated lesion detection of diabetic retinopathy using morphological operation. Epub ahead of print 13 November 2020. Signal Image and Video Processing.
- [5] Sudha V, Ganeshbabu TR. 2021. A convolutional neural network classifier VGG-19 architecture for lesion detection and grading in diabetic retinopathy based on deep learning. Computers, Materials & Continua 66(1):827–842.
- [6] Vaishnavi J, Ravi S, Anbarasi A. 2020. An efficient adaptive histogram based segmentation and extraction model for the classification of severities on diabetic retinopathy. Multimedia Tools and Applications.

Future Work

The obtained results in the research show that image processing techniques combined with optimization methods can still be competitive to convolutional network and deep learning-based approaches. The proposed method also has some limitations as using all textural and Ridgelet features may include irrelevant features for the task of DR recognition, which can incur larger computation time, and sometimes even reduce the recognition accuracy. These limitations could be overcome by further fine-tuning the parameters of SMO technique.

Presented By

Faisal Farhan

Erasmus Mundus Joint Master's degree in Medical Imaging and Applications (MAIA)

